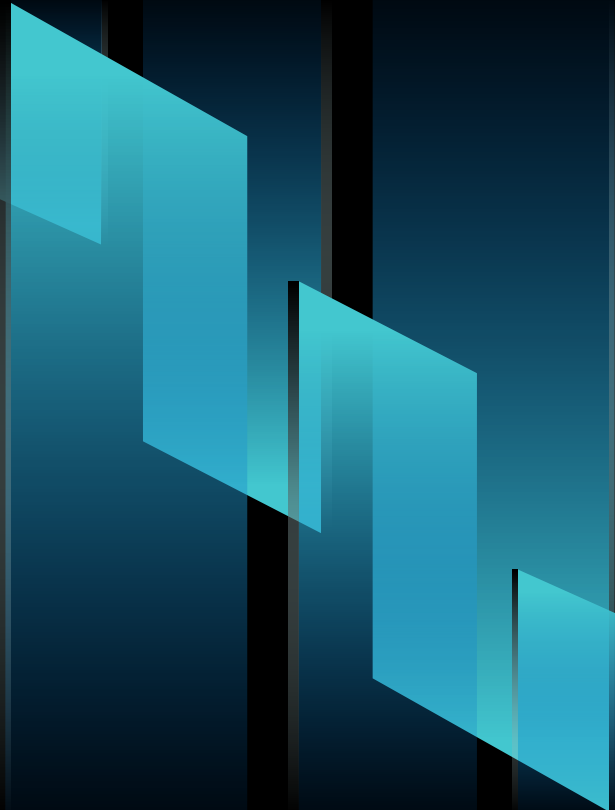


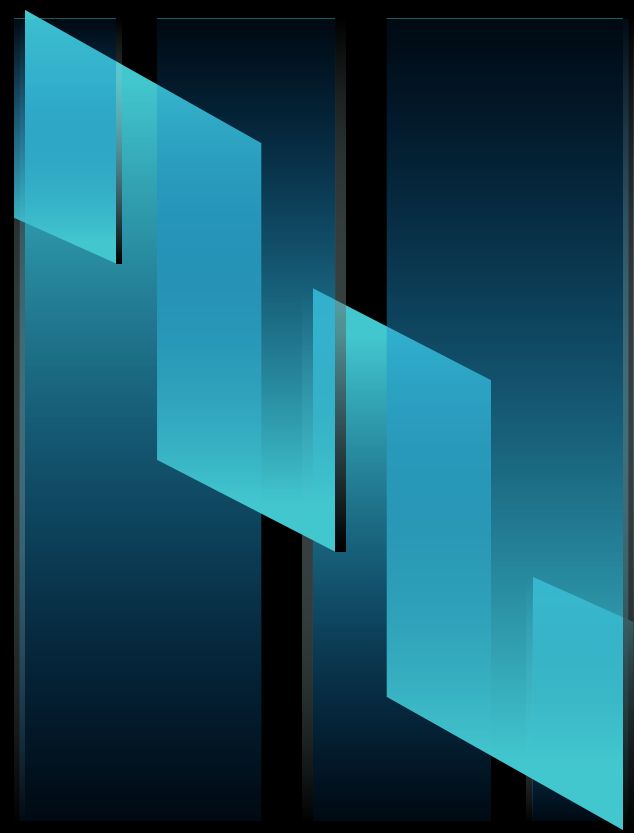


TELECOM CUSTOMER CHURN ANALYSIS USING PYTHON



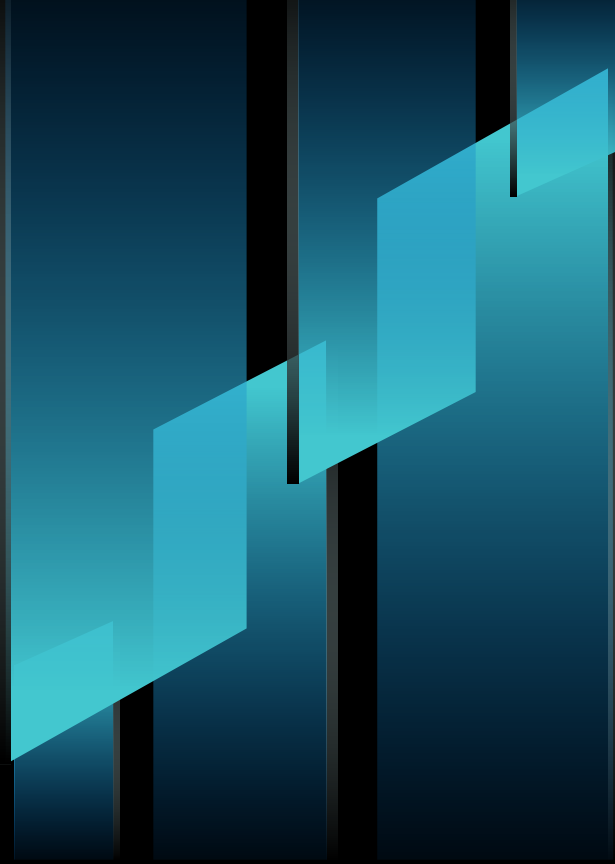
WHAT IS CHURN ANALYSIS

Customer churn analysis refers to identifying customers who are likely to stop using a company's product or service. It helps in :

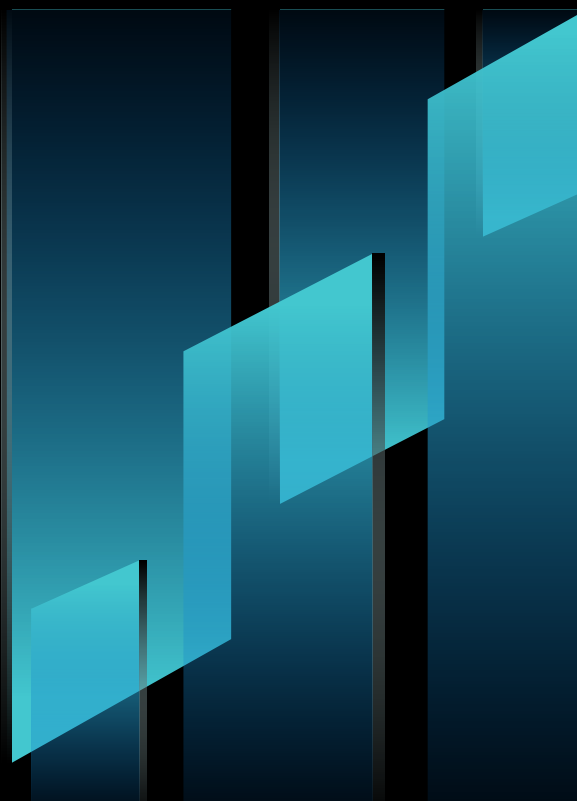
- Understanding why customers leave.
 - Predicting future churn.
 - Creating retention strategies.
- 

TOOLS & TECHNOLOGIES USED

- Jupyter Notebook: To execute code and display results.
- Python: A programming language for data analysis.
- Libraries Used: Numpy, Pandas, Matplotlib, and Seaborn.
- CSV Dataset: "Customer Churn.csv"



BUSINESS REQUIREMENTS

1. What percentage of customers have churned?
 2. Is churn related to gender or age group (senior citizen)?
 3. How does customer tenure affect churn?
 4. Which contract type leads to higher churn?
 5. Does the use of specific services (e.g., online security, tech support) impact churn?
 6. Do payment methods influence churn rate?
 7. Are there data quality issues that need cleaning before analysis?
- 

IMPORTING REQUIRED LIBRARIES

```
▶ import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns
```

“We import essential Python libraries for data loading, manipulation, and visualization.”

LOADING THE DATASET AND DISPLAYING INITIAL ROWS

```
path="/content/Customer Churn.csv"
df = pd.read_csv(path)
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	Internet
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	

“We load the churn dataset into a pandas DataFrame to begin our analysis and preview the first few records to understand the structure and data types.”

INSPECTING DATA INFORMATION

```
df.info()

... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column              Non-Null Count  Dtype
---  -
 0   customerID          7043 non-null   object
 1   gender              7043 non-null   object
 2   SeniorCitizen       7043 non-null   int64
 3   Partner             7043 non-null   object
 4   Dependents          7043 non-null   object
 5   tenure              7043 non-null   int64
 6   PhoneService        7043 non-null   object
 7   MultipleLines       7043 non-null   object
 8   InternetService     7043 non-null   object
 9   OnlineSecurity      7043 non-null   object
10  OnlineBackup        7043 non-null   object
11  DeviceProtection    7043 non-null   object
12  TechSupport         7043 non-null   object
```

“Checking column data types, non-null values, and overall structure of the dataset.”

HANDLING NULL AND INCONSISTENT DATA

Replacing blanks with 0 as tenure is 0 and total charges are recorded and converted the datatype from object to float

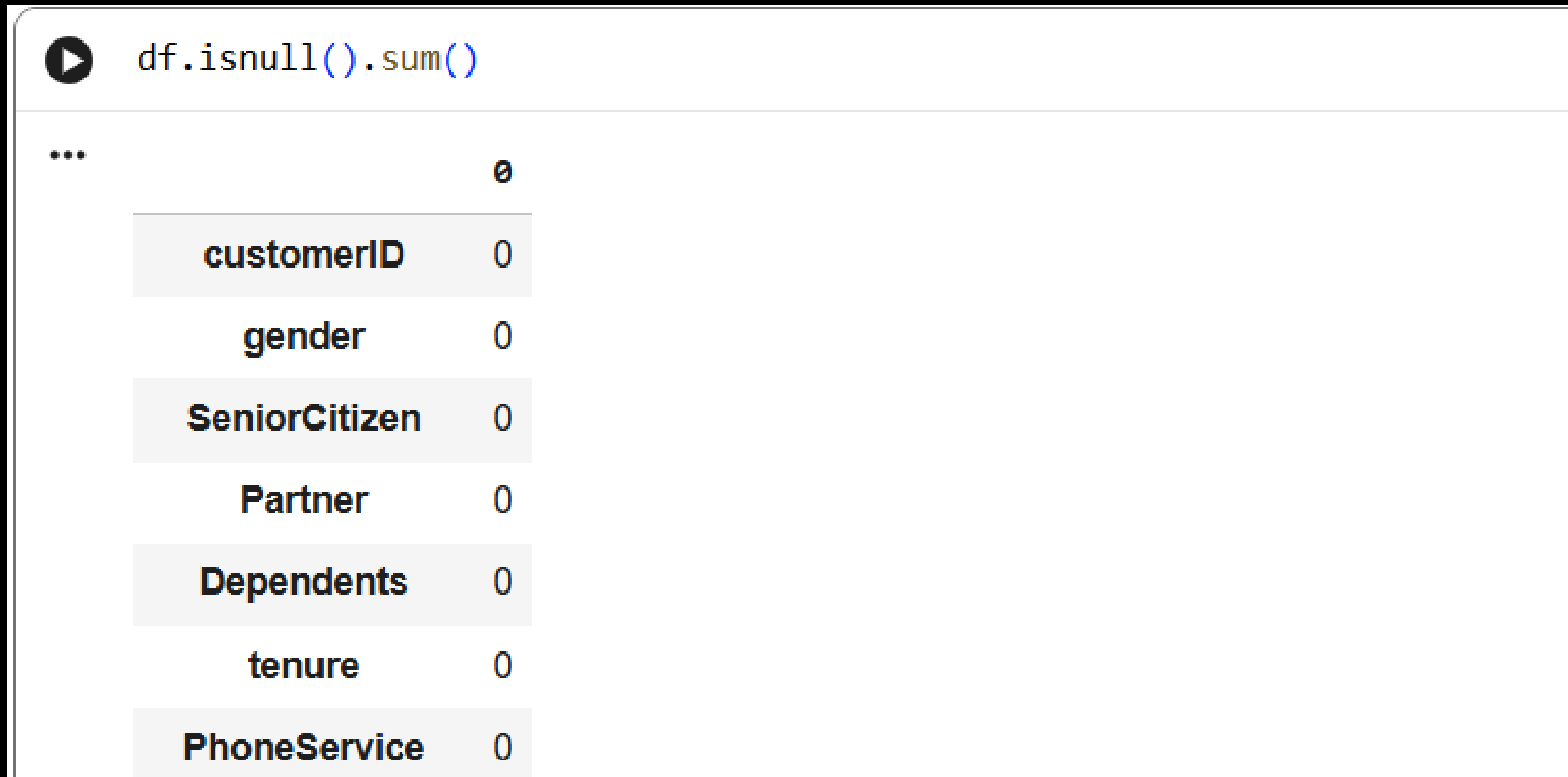
```
df["TotalCharges"] = df["TotalCharges"].replace(" ", "0")  
df["TotalCharges"] = df["TotalCharges"].astype("float")
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 7043 entries, 0 to 7042  
Data columns (total 21 columns):  
#   Column                Non-Null Count  Dtype  
---  -  
0   customerID            7043 non-null   object  
1   gender                 7043 non-null   object  
2   SeniorCitizen          7043 non-null   int64  
3   Partner                7043 non-null   object  
4   Dependents             7043 non-null   object  
5   tenure                 7043 non-null   int64
```

“Fixing missing or incorrectly formatted values in the ‘TotalCharges’ column.”

CHECKING FOR NULL VALUES



A screenshot of a Jupyter Notebook interface. The top bar shows a play button icon and the code `df.isnull().sum()`. Below the code, the output is displayed as a table with two columns: the column name and the count of null values. The output shows that all columns have zero null values.

...	0
customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0

“Fixing missing or incorrectly formatted values in the ‘TotalCharges’ column.”

CHECKING FOR DUPLICATE RECORDS

```
▶ df.duplicated().sum()
```

```
... np.int64(0)
```

```
df["customerID"].duplicated().sum()
```

```
np.int64(0)
```

“Verifying that customer records are unique and there are no duplicates.”

DESCRIPTIVE STATISTICS

df.describe()

...	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2266.794470
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

“Getting summary statistics (mean, std, min, max) for numeric columns.”

CONVERTING NUMERIC TO CATEGORICAL

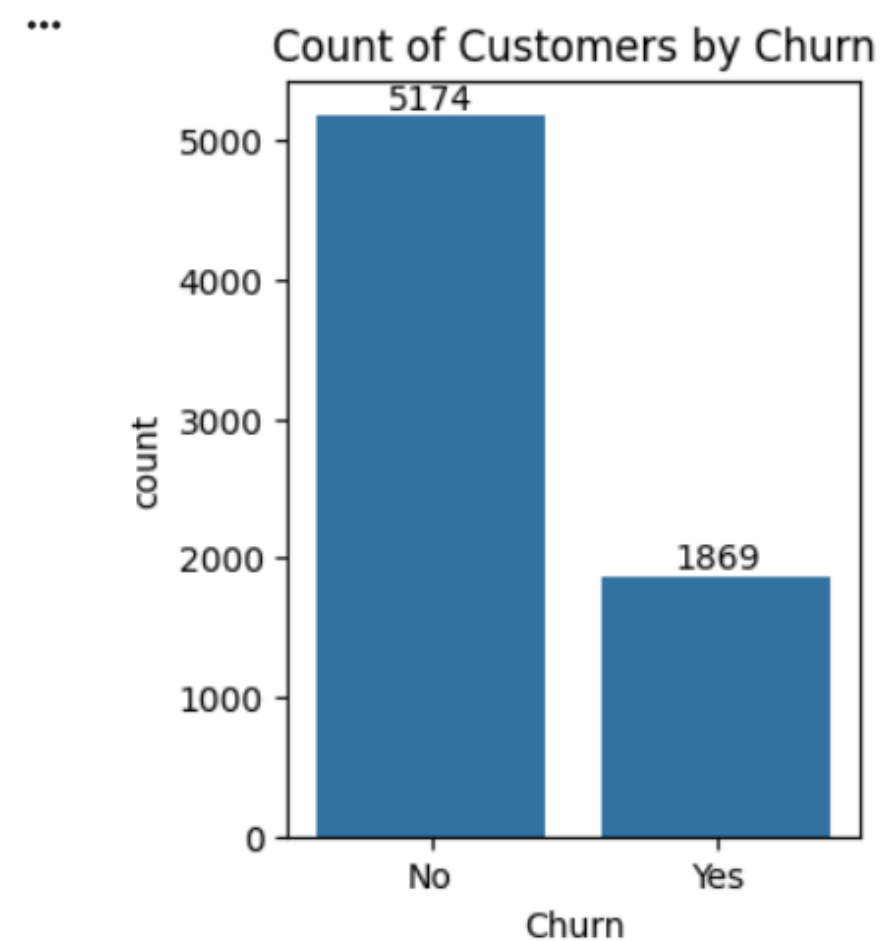
The "SeniorCitizen" column is given as 1 and 0 converted it to yes and no to make it easy to understand

```
▶ def conv(value):  
    if value == 1:  
        return "yes"  
    else:  
        return "no"  
  
df["SeniorCitizen"] = df["SeniorCitizen"].apply(conv)
```

“Making ‘SeniorCitizen’ easier to interpret by converting numeric to 'Yes/No'.”

COUNT OF CHURNED CUSTOMERS

```
plt.figure(figsize=(3,4))  
ax = sns.countplot(x = "Churn", data = df)  
ax.bar_label(ax.containers[0])  
plt.title("Count of Customers by Churn")  
plt.show()
```

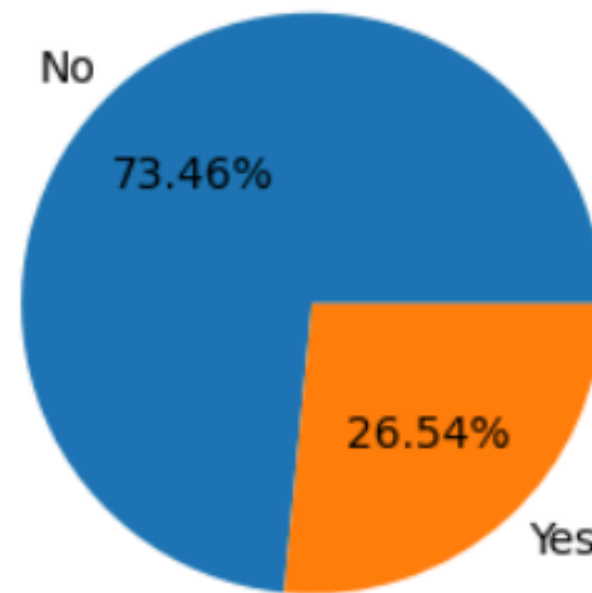


“Visualizing how many customers have churned vs. retained.”

CHURN PERCENTAGE

```
plt.figure(figsize = (3, 4))  
gb = df.groupby("Churn").agg({"Churn": "count"})  
plt.pie(gb["Churn"], labels = gb.index, autopct = "%1.2f%%")  
plt.title("Percentage of Churned Customers", fontsize = 10)  
plt.show()
```

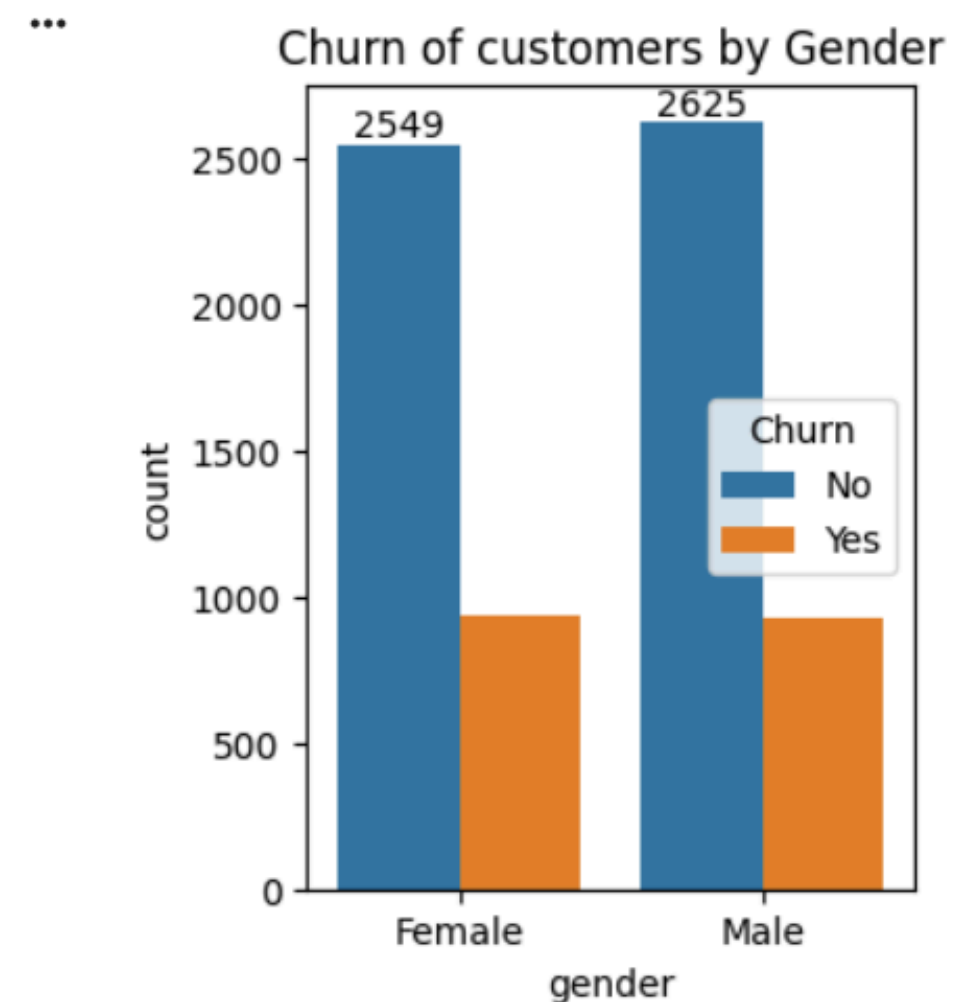
... Percentage of Churned Customers



“Visualizing the percentage of churned customers using a pie chart.”

CHURN BY GENDER

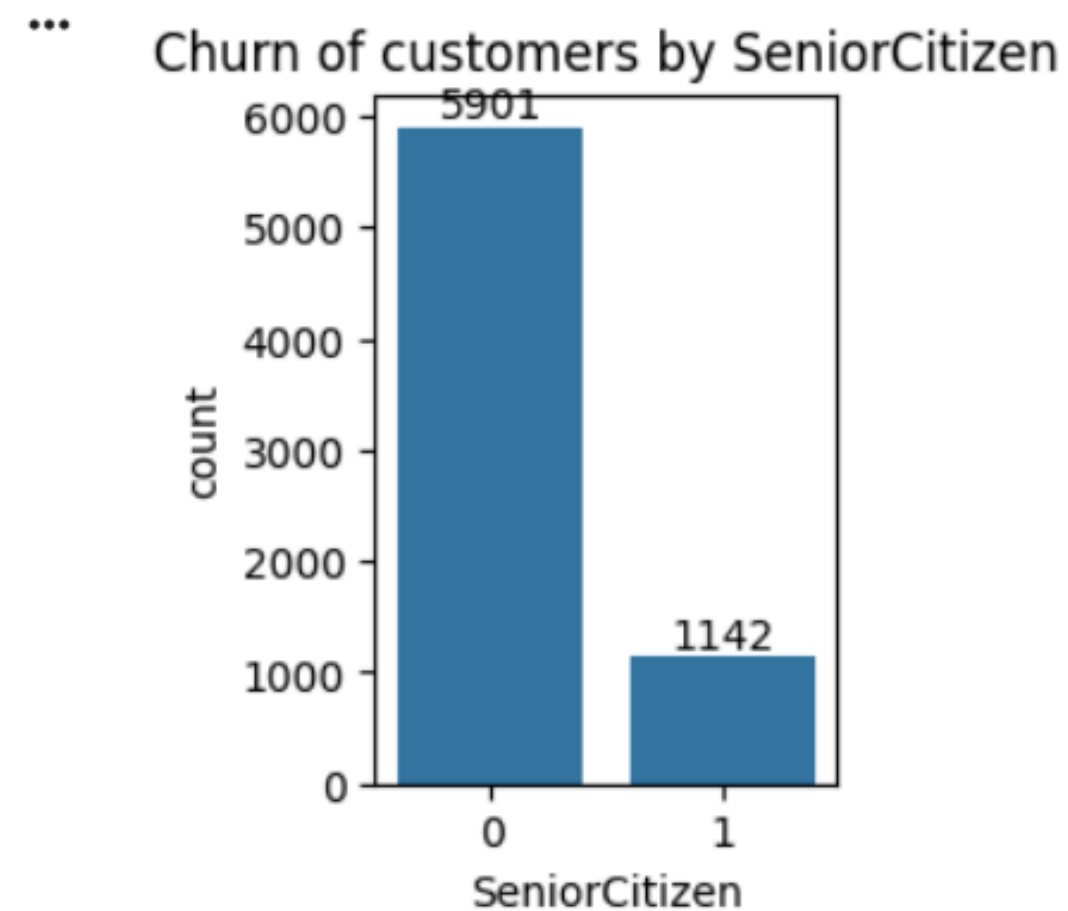
```
plt.figure(figsize=(3,4))  
ax = sns.countplot(x = "gender", data = df, hue = "Churn")  
plt.title("Churn of customers by Gender")  
ax.bar_label(ax.containers[0])  
plt.show()
```



“Analyzing if churn behavior varies by gender.”

COUNT OF SENIOR CITIZEN

```
plt.figure(figsize=(2,3))  
ax = sns.countplot(x = "SeniorCitizen", data = df)  
plt.title("Churn of customers by SeniorCitizen")  
ax.bar_label(ax.containers[0])  
plt.show()
```



“Exploring how many senior citizen customers we have.”

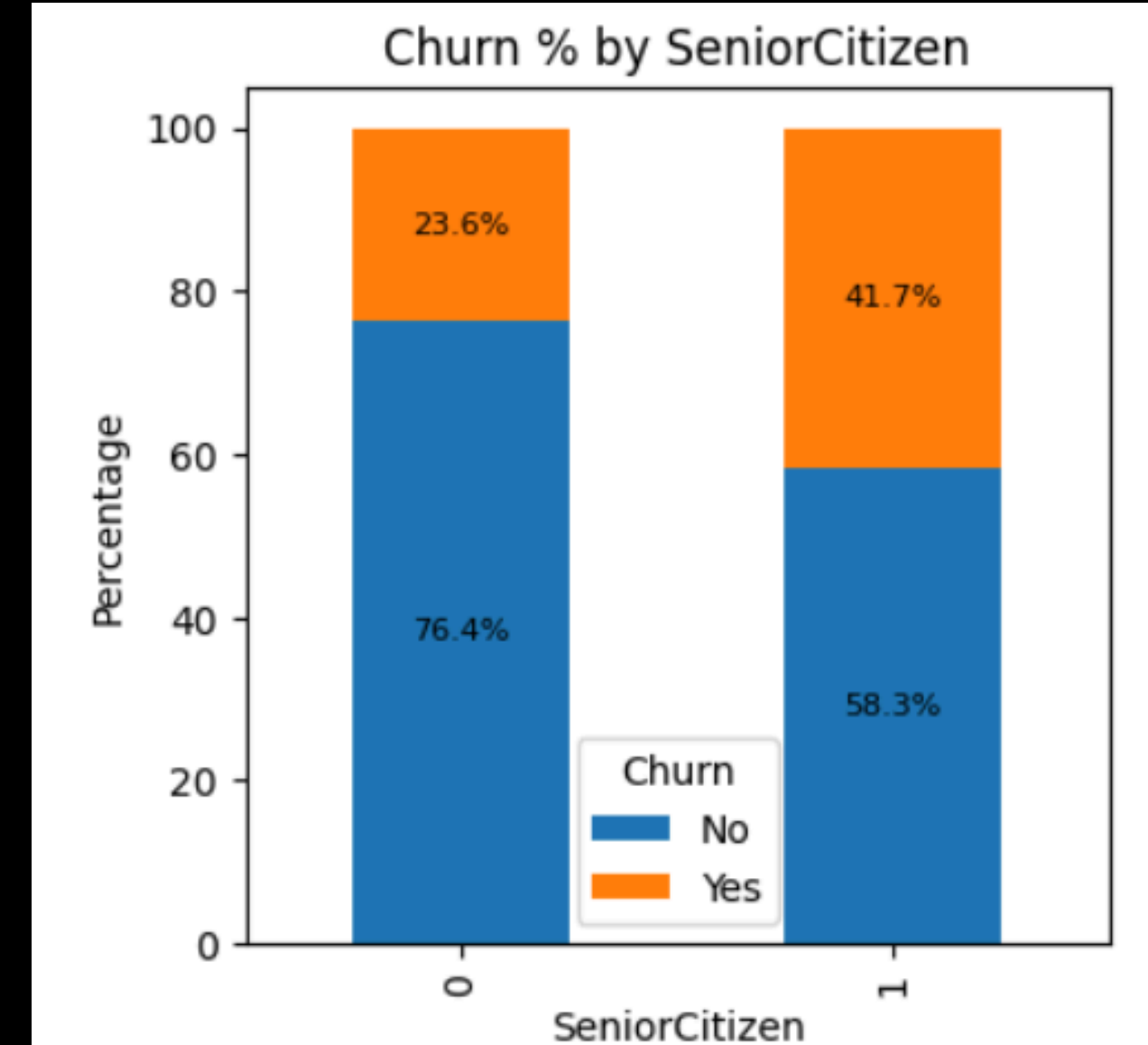
CHURN ANALYSIS BY SENIOR CITIZEN

```
plt.figure(figsize=(3,3))

# create crosstab
ct = pd.crosstab(df['SeniorCitizen'], df['Churn'], normalize='index') * 100
ct.plot(kind='bar', stacked=True, figsize=(4,4))

# add % labels
for i, row in ct.iterrows():
    cum = 0
    for churn, val in row.items():
        plt.text(i, cum + val/2, f"{val:.1f}%", ha='center', va='center', fontsize=8)
        cum += val

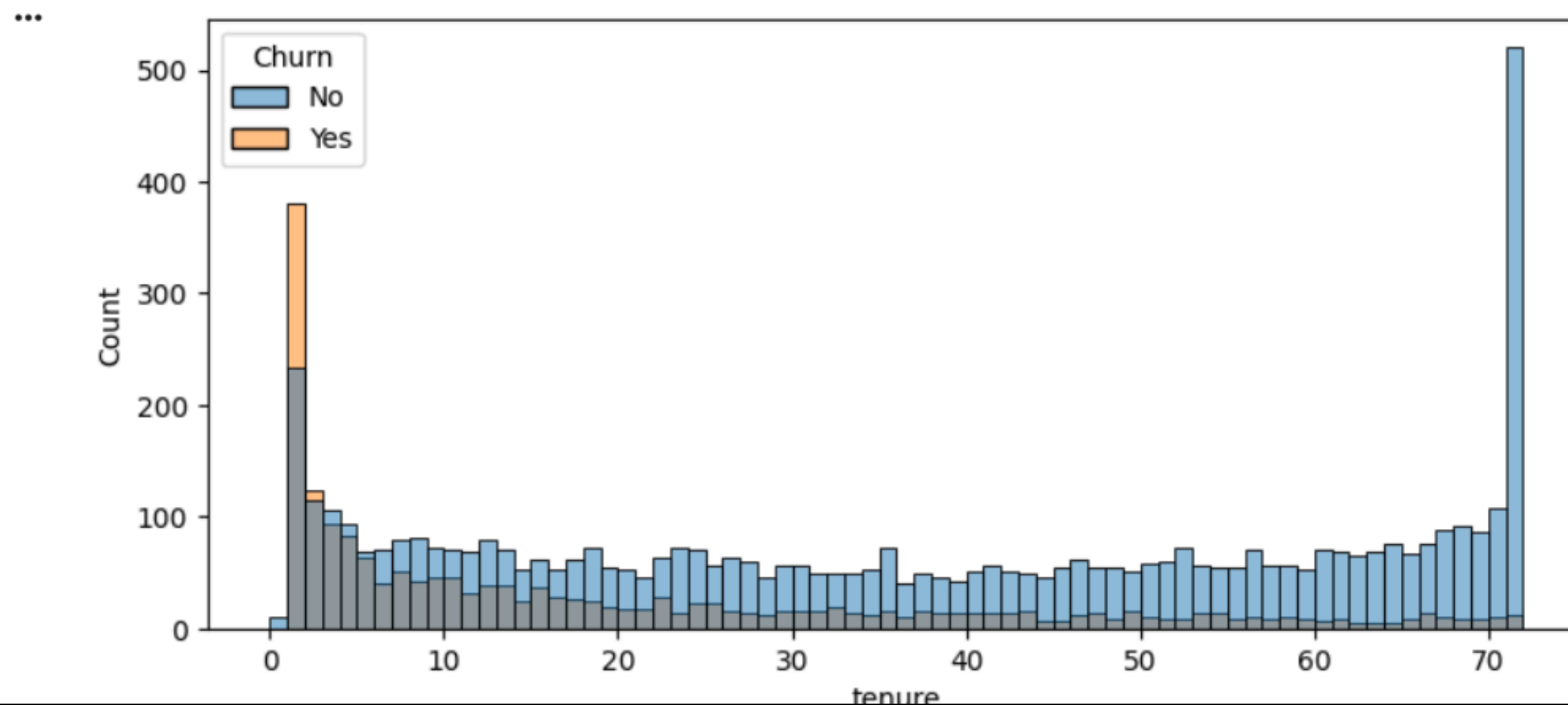
plt.title("Churn % by SeniorCitizen")
plt.ylabel("Percentage")
plt.legend(title="Churn")
plt.show()
```



“Stacked bar showing churn percentage among senior and non-senior citizens for comparison.”

CHURN VS. TENURE

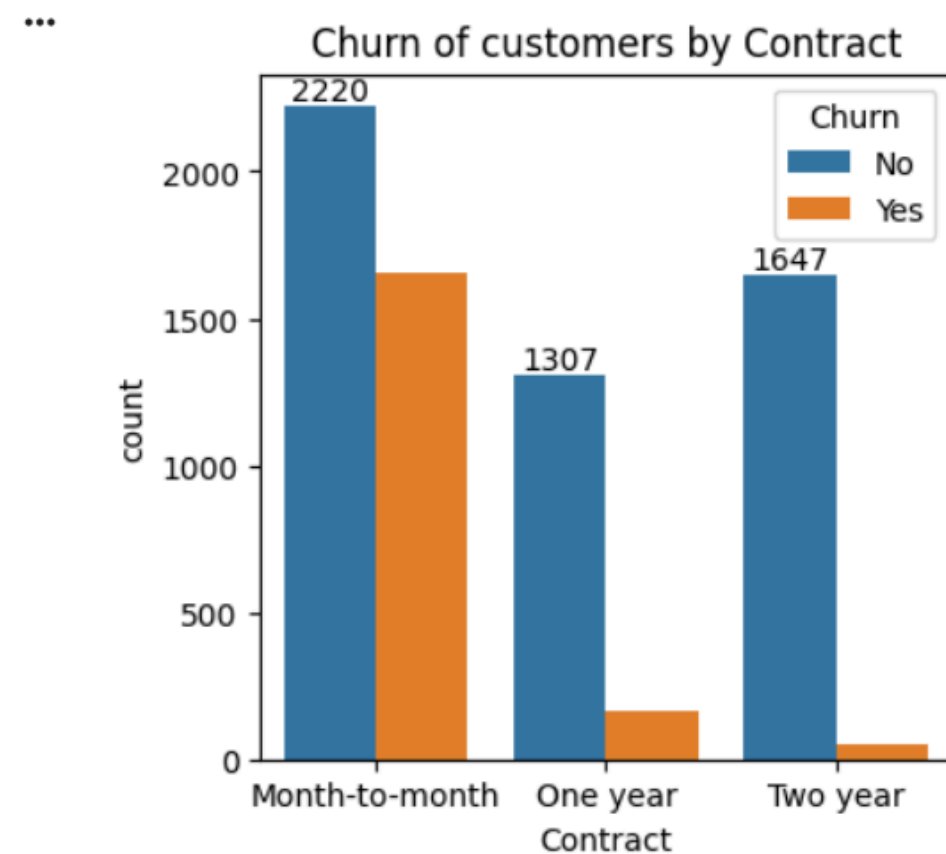
```
plt.figure(figsize = (9,4))  
sns.histplot(x = "tenure", data = df, bins = 72, hue = "Churn")  
plt.show()
```



“Understanding how contract duration impacts churn.”

CHURN BY CONTRACT TYPE

```
plt.figure(figsize=(4,4))  
ax = sns.countplot(x = "Contract", data = df, hue = "Churn")  
plt.title("Churn of customers by Contract")  
ax.bar_label(ax.containers[0])  
plt.show()
```



“Churn analysis by contract type to see which contract leads to higher customer loss.

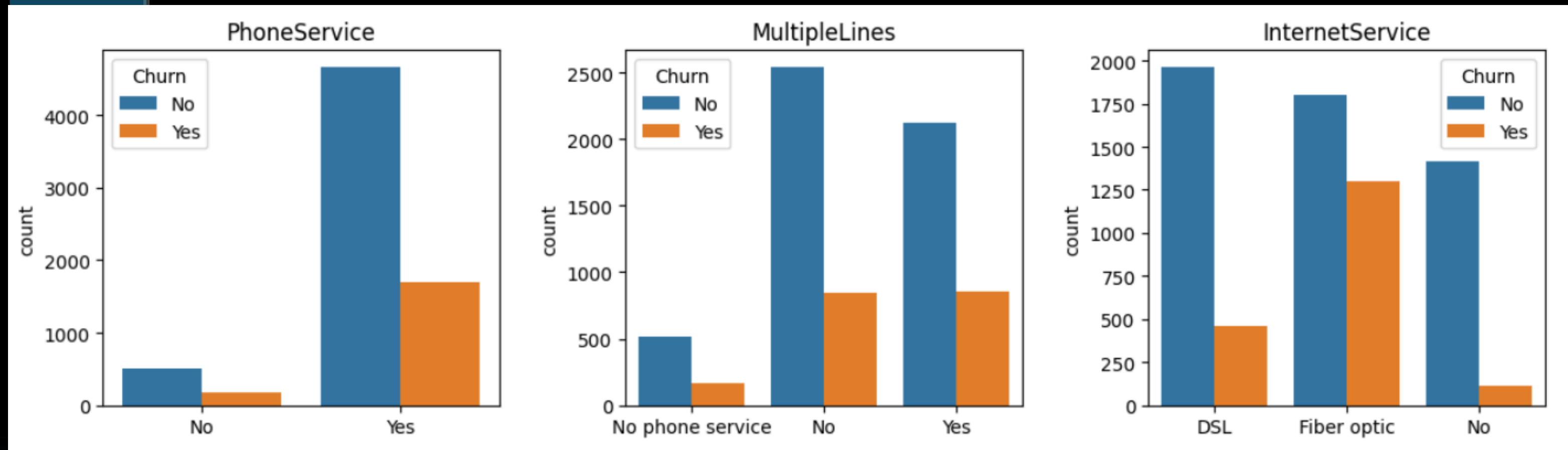
CHURN BY AVAILABLE SERVICES



```
cols = ['PhoneService', 'MultipleLines', 'InternetService',  
        'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',  
        'TechSupport', 'StreamingTV', 'StreamingMovies']  
  
plt.figure(figsize=(12,10))  
  
for i, col in enumerate(cols, 1):  
    plt.subplot(3, 3, i)  
    sns.countplot(x=df[col], hue = df["Churn"])  
    plt.title(col)  
    plt.xlabel("")  
    plt.tight_layout()  
  
plt.show()
```

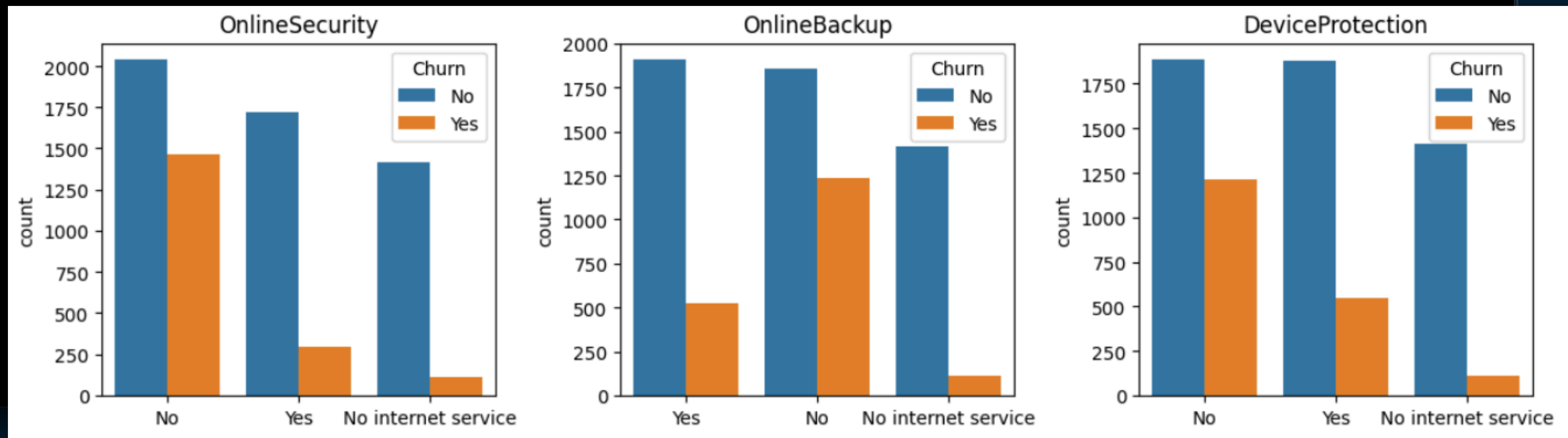
“Churn comparison across various customer services to identify which services are linked to higher churn rates.”

CHURN BY AVAILABLE SERVICES



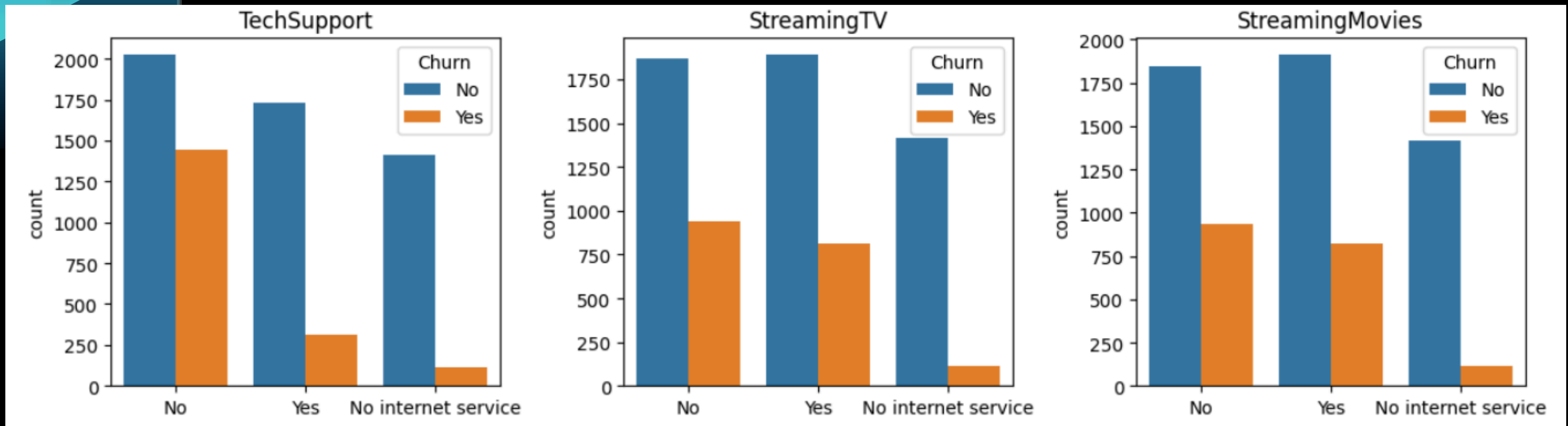
“Churn comparison across various customer services to identify which services are linked to higher churn rates.”

CHURN BY AVAILABLE SERVICES



“Churn comparison across various customer services to identify which services are linked to higher churn rates.”

CHURN BY AVAILABLE SERVICES



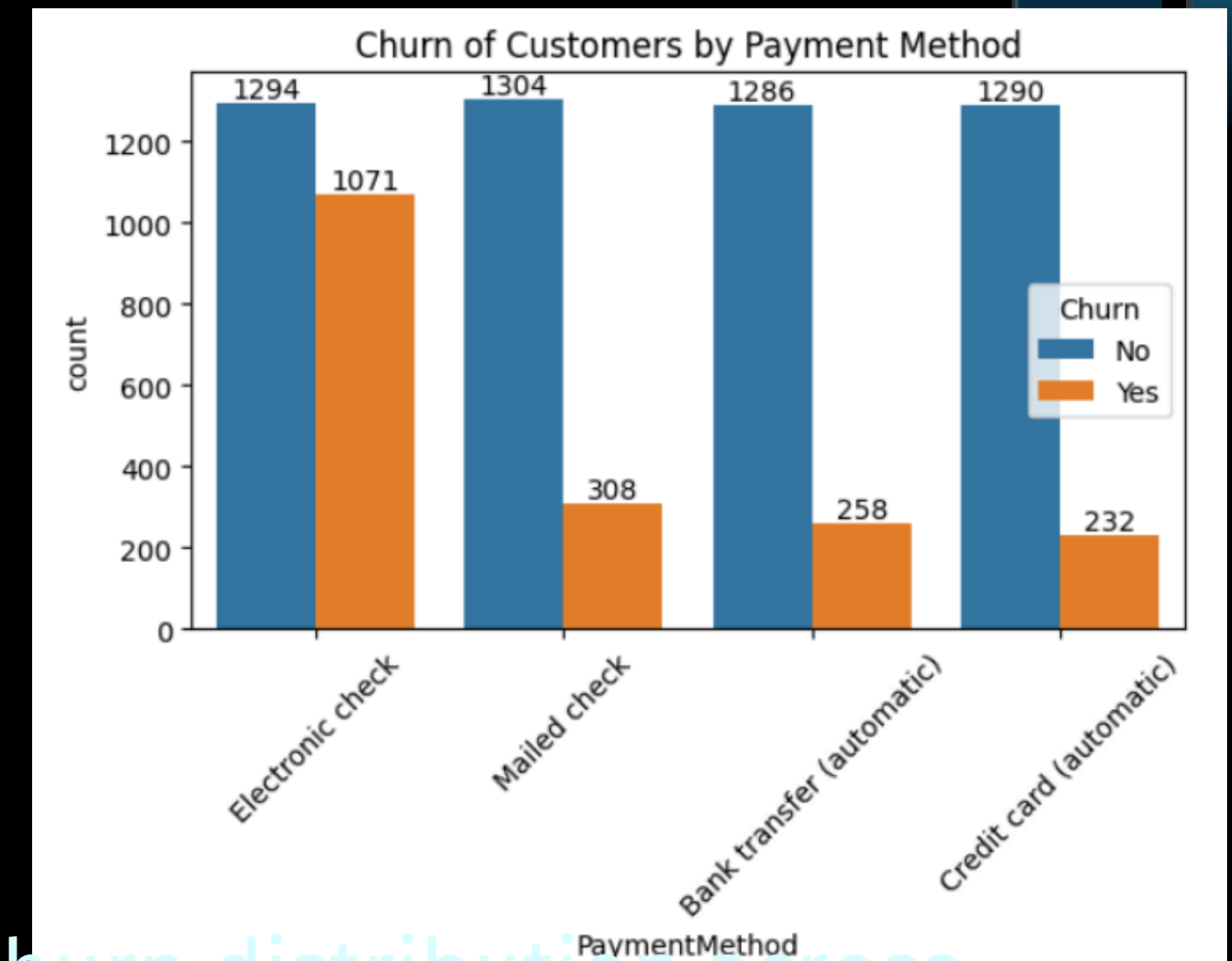
“Churn comparison across various customer services to identify which services are linked to higher churn rates.”

CHURN BY PAYMENT METHOD

```
plt.figure(figsize=(6,5))
ax = sns.countplot(x="PaymentMethod", data=df, hue="Churn")

ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])

plt.title("Churn of Customers by Payment Method")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



“Grouped bar chart showing churn distribution across different payment methods.”

KEY FINDINGS FOR CUSTOMER RETENTION

1. High Churn: A 26.54% churn rate (1,869 customers) demands urgent attention.
2. Gender-Neutral Churn: Churn rates are consistent across genders, suggesting the need for universal retention efforts.
3. Senior Citizen Loyalty: Senior citizens churn less, indicating effective engagement strategies in this demographic.
4. Contract Vulnerability: Month-to-month contracts carry the highest risk of churn; long-term contracts significantly reduce churn.
5. Service Gaps & Churn: Customers are more likely to churn if they lack extra services (like online security or tech help) or if they have fiber-optic internet.
6. Payment Method Risk: Electronic Check users show a higher churn rate, requiring investigation.

RECOMMENDATIONS

1. Prioritize Contract Type: Focus retention efforts on customers on month-to-month contracts by offering incentives for longer-term commitments.
2. Enhance Service Bundling: Promote the adoption of value-added services (OnlineSecurity, TechSupport, etc.) to increase customer stickiness and perceived value.
3. Investigate Fiber Optic Churn: Conduct deeper analysis into Fiber Optic customer feedback and service quality to identify and address pain points.
4. Optimize Electronic Check Experience: Analyze the reasons behind higher churn among Electronic Check users; consider alternative payment method promotions or improving the electronic check process.
5. Leverage Senior Citizen Success: Understand what makes senior citizens less likely to churn and explore if these strategies can be applied to other demographics.

The background is a dark blue gradient. On the left and right sides, there are vertical panels. Each panel features a grid of small, light blue dots. Overlaid on these panels are several translucent, light blue rectangular blocks of varying heights and widths, arranged in a staggered, ascending pattern from bottom-left to top-right. The central text is white and bold.

THANK
YOU